

Patient: Kimberley smith

ID: 70 | Email: kimberlycristena@gmail.com

DOB: 2000-01-21 | Gender: female

Report Type: Comprehensive

Tests Completed: 30

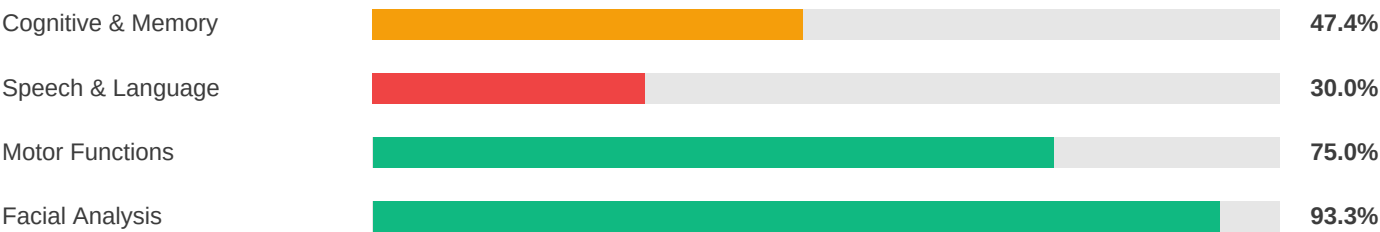
Assessed by: Dr. Muhammad Naeem

1. Overall Risk Assessment



AD Stage: Normal | PD Stage: Unilateral involvement only

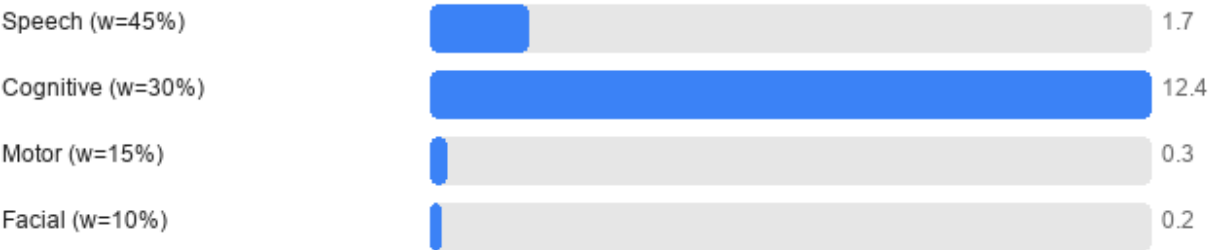
2. Category-Wise Health Scores



3. Fusion Weight Breakdown

Shows how each test category contributes to the final AD and PD risk scores. Weights reflect clinical evidence for early detection sensitivity.

AD Fusion Contributions



PD Fusion Contributions



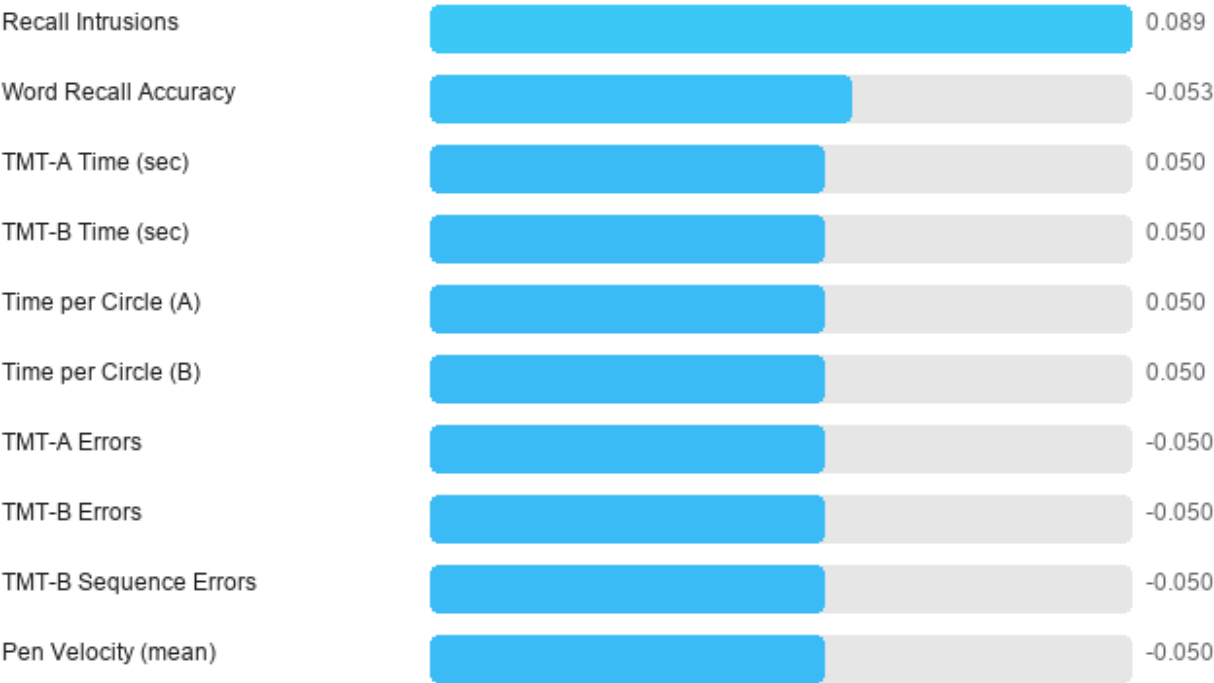
4. Cognitive & Memory - Detailed Analysis

Health Score: 47.4% | AD Risk: 41.3% | PD Risk: 7.2% | Stage: Moderate Dementia | Severity: high

SHAP Feature Attribution

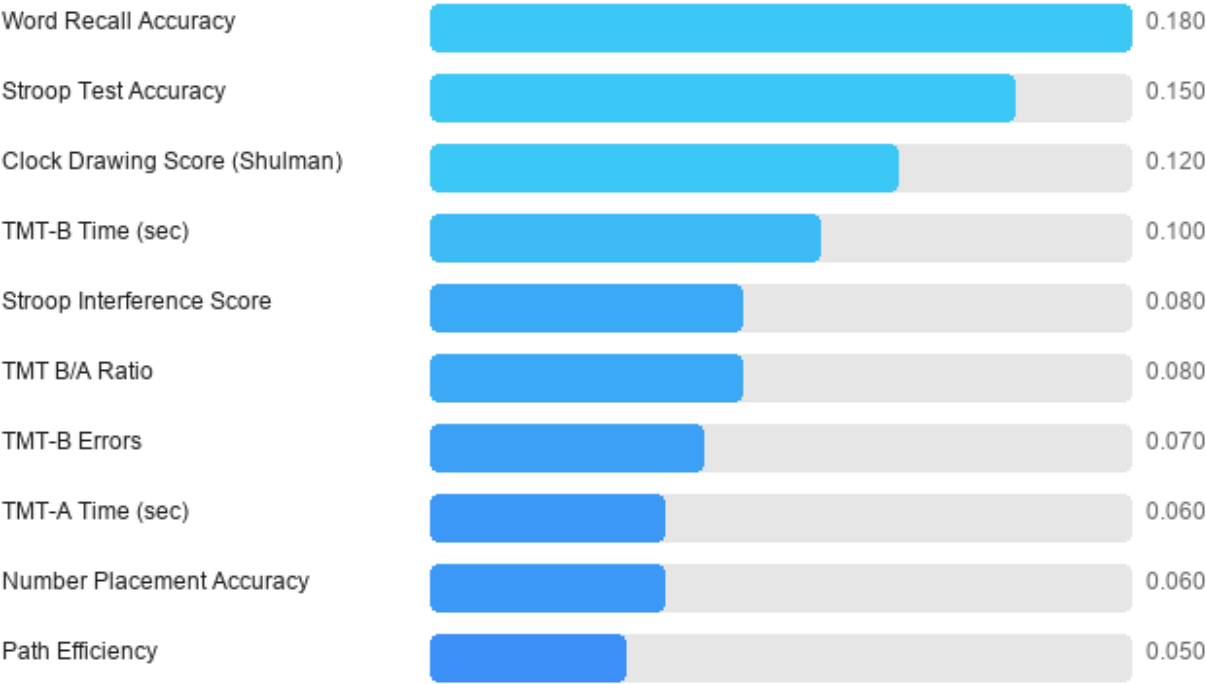
SHAP values show each feature's contribution to the risk prediction. Positive values increase risk; negative values are protective.

SHAP Feature Importance



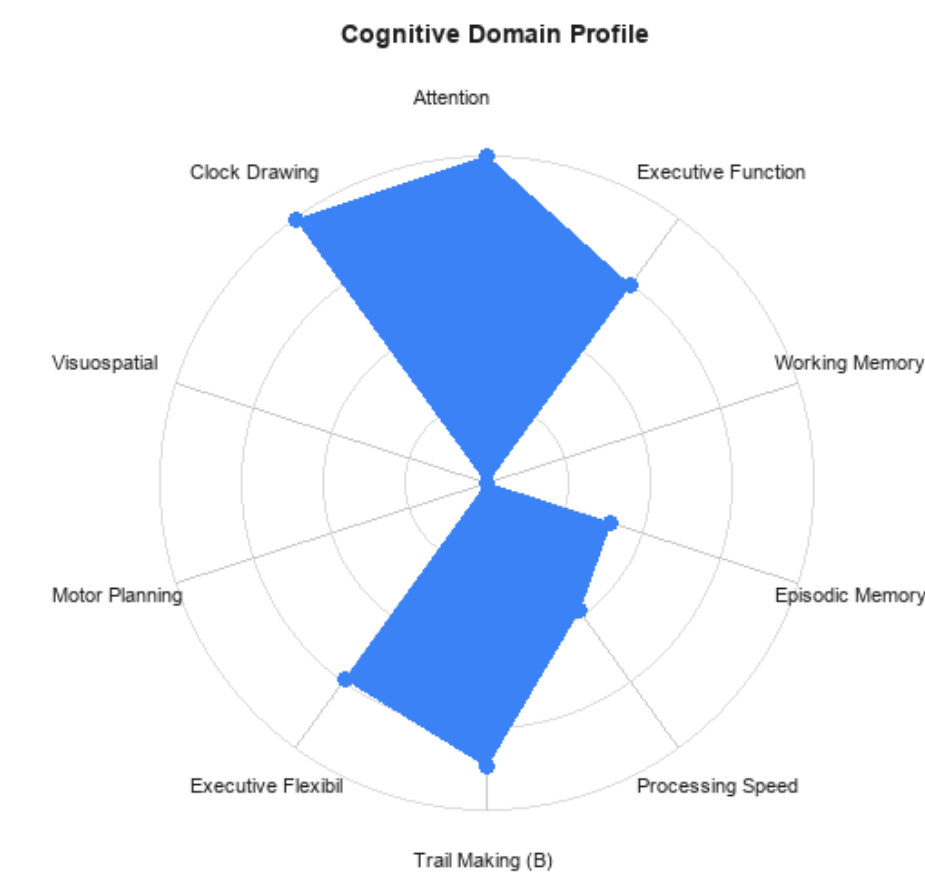
Feature Importance Ranking

Top Features by Importance



Cognitive Domain Profile (Radar Chart)

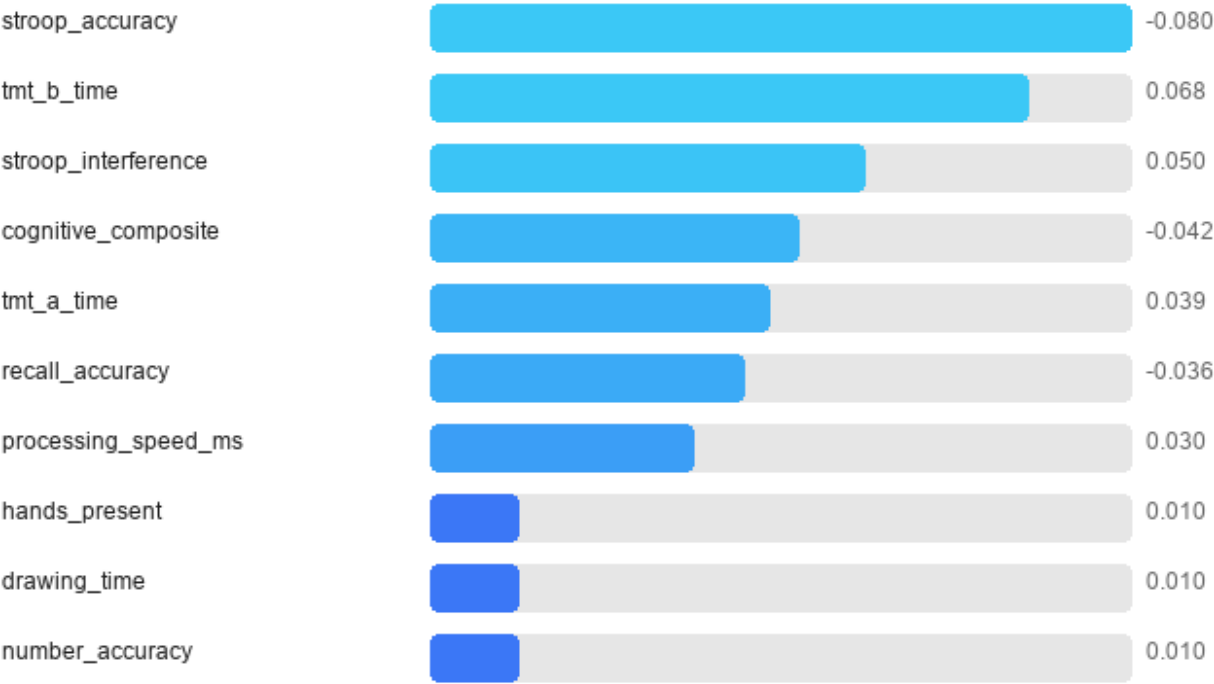
Scores represent performance across cognitive domains. Values closer to the outer edge indicate better performance.



LIME - Local Interpretable Explanation

LIME identifies which features most influenced this specific prediction by approximating the model's decision boundary locally.

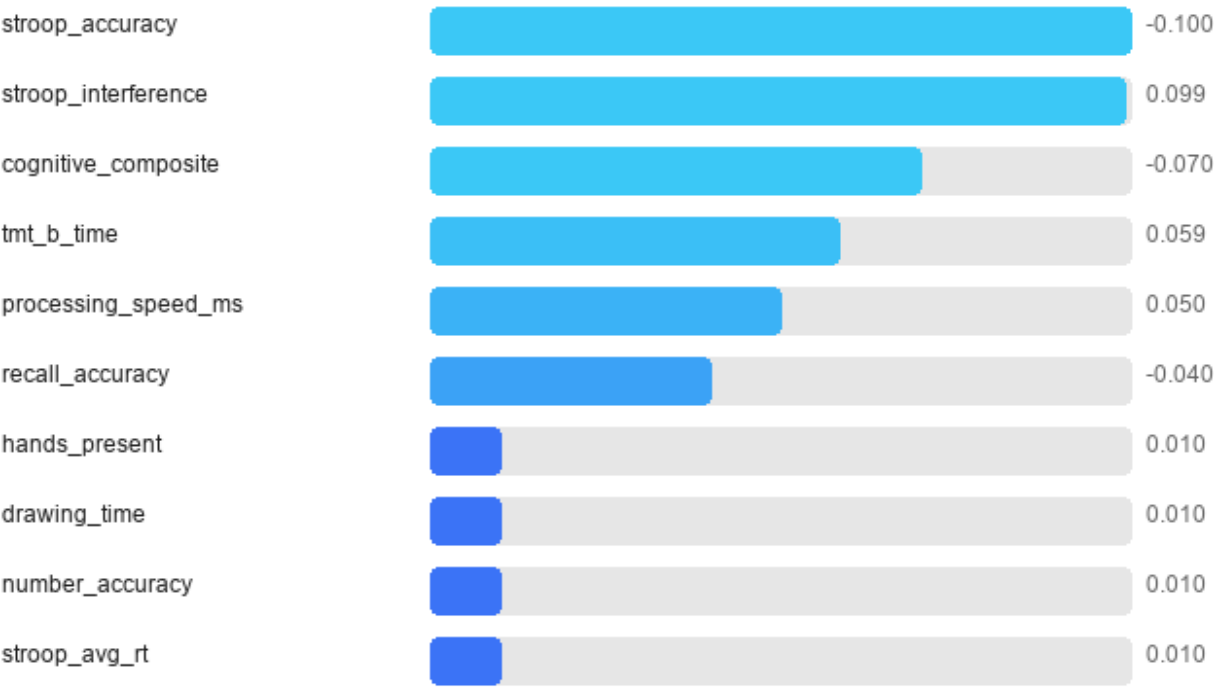
LIME Feature Weights



Integrated Gradients - Path-Integrated Attribution

Shows attribution accumulated along the path from a baseline to the actual input, providing a complete picture of each feature's contribution.

Integrated Gradient Attributions



Counterfactual Analysis - 'What-If' Scenarios

Attention Pattern Visualization

Shows where the neural network focused its attention when processing the input data.

Clinical Interpretation

Moderate AD Risk Indicators

Your cognitive assessment shows some patterns (41% risk) that warrant monitoring over time.

TMT-A Time (sec) - Below Expected

TMT-A Time (sec) is 30.60, significantly below the normal range (20-60).

TMT-B Time (sec) - Below Expected

TMT-B Time (sec) is 39.37, significantly below the normal range (40-120).

TMT-B Errors - Below Expected

TMT-B Errors is 0.00, significantly below the normal range (0-1).

Pen Lifts - Below Expected

Pen Lifts is 0.00, significantly below the normal range (0-5).

Path Efficiency - Below Expected

Path Efficiency is 0.00, significantly below the normal range (0.7-1.0).

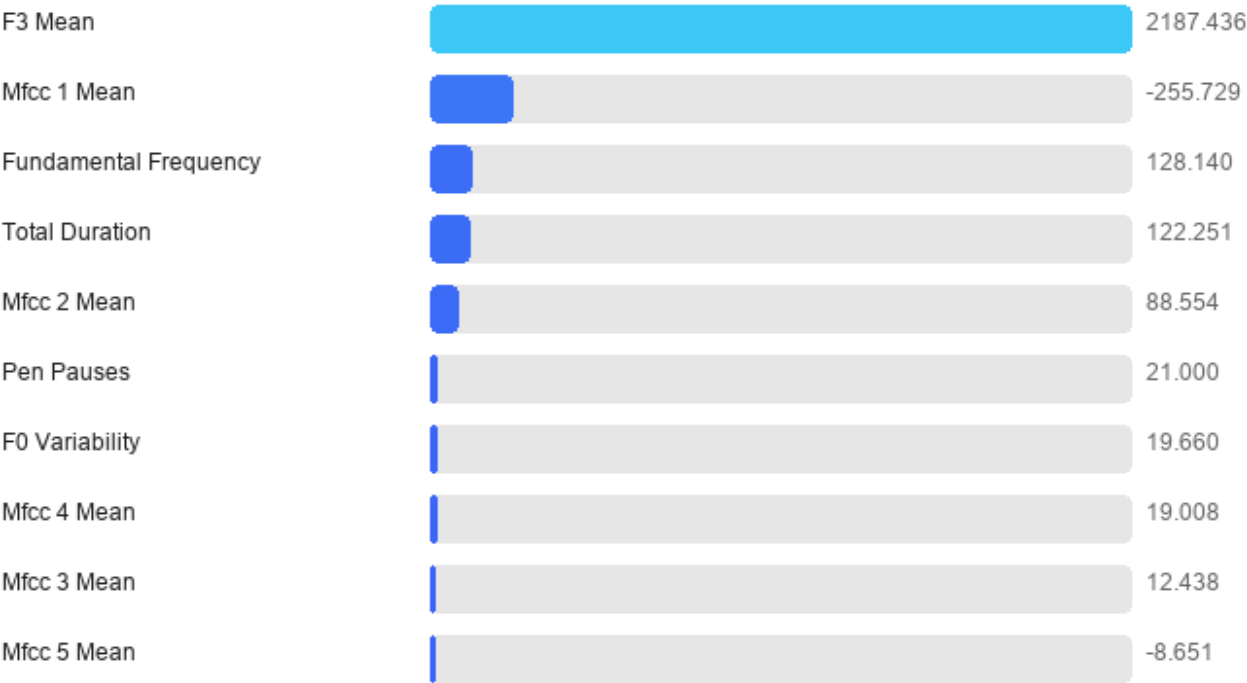
5. Speech & Language - Detailed Analysis

Health Score: 30.0% | AD Risk: 3.8% | PD Risk: 6.8% | Stage: Minimal | Severity: mild

SHAP Feature Attribution

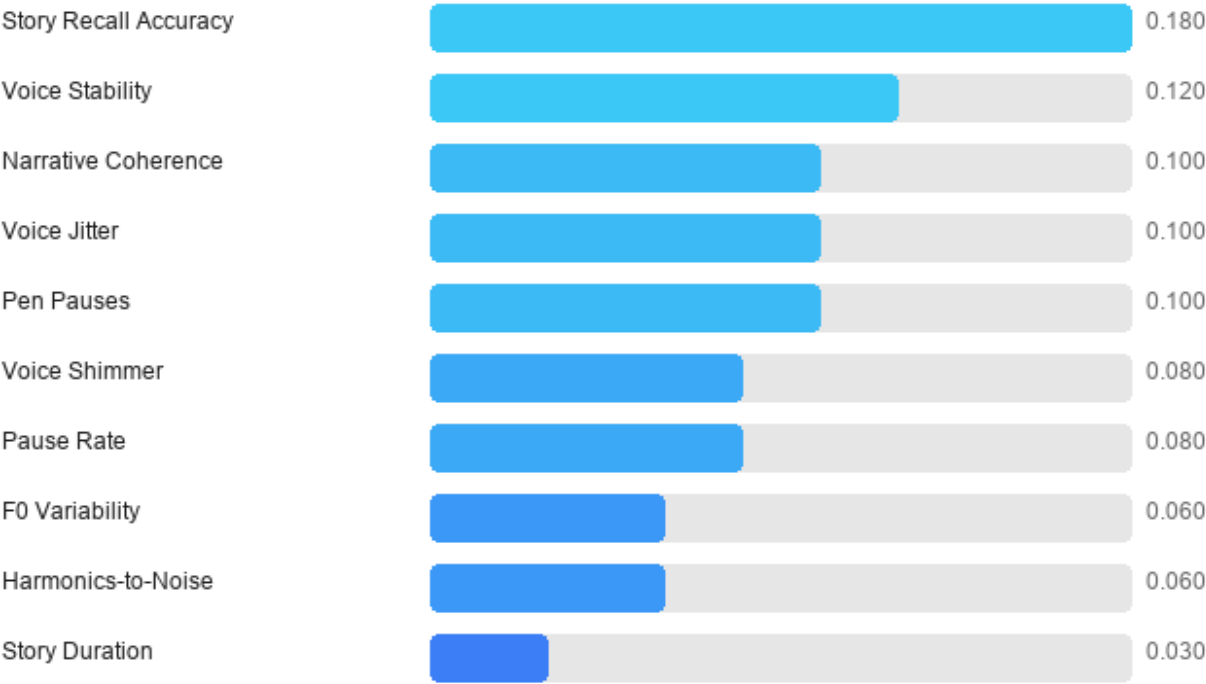
SHAP values show each feature's contribution to the risk prediction. Positive values increase risk; negative values are protective.

SHAP Feature Importance



Feature Importance Ranking

Top Features by Importance



Feature Analysis Bars

Speech & Language Features



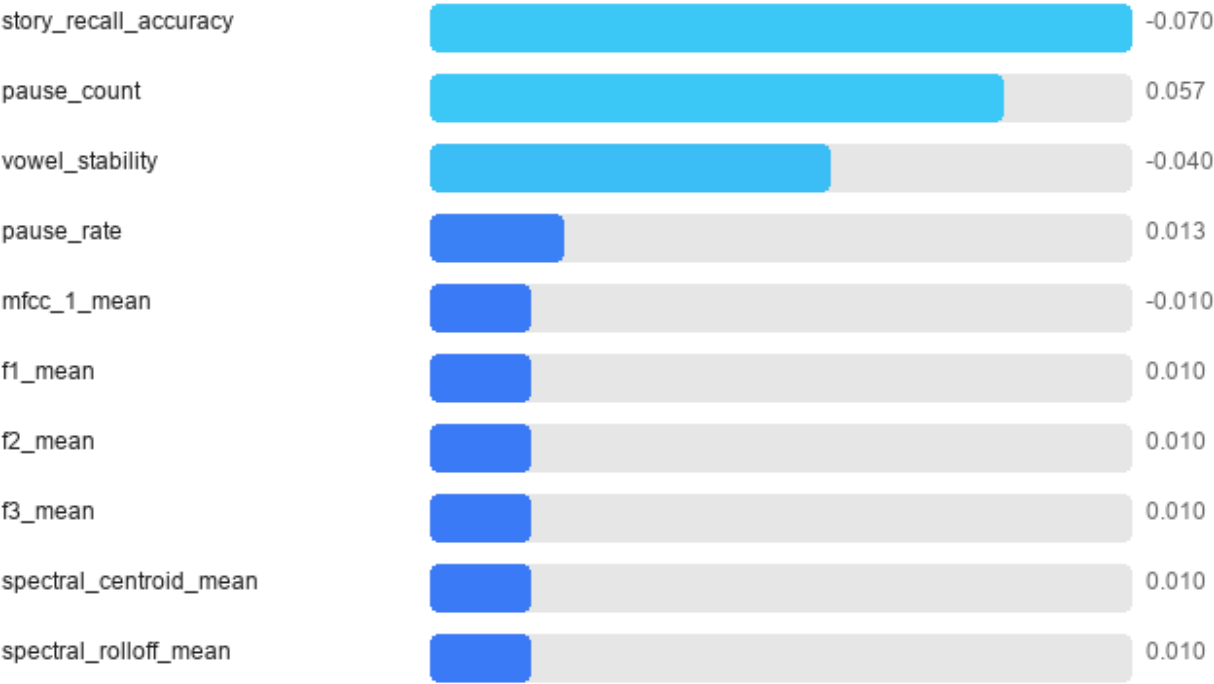
Clinical Highlights

- Excessive pauses detected – potential word-finding difficulty
- Voice irregularity detected – potential motor speech issue

LIME - Local Interpretable Explanation

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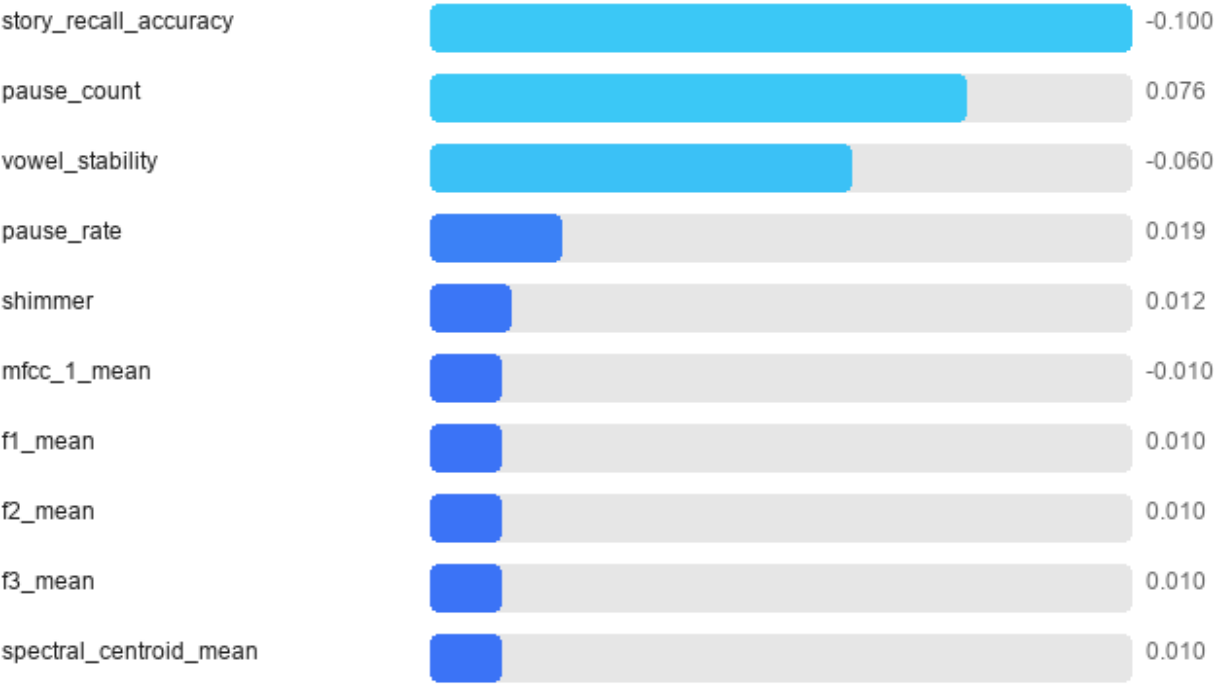
LIME Feature Weights



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Integrated Gradient Attributions



Counterfactual Analysis - 'What-If' Scenarios

Attention Pattern Visualization

Shows where the neural network focused its attention when processing the input data.

Attention Weights



Clinical Interpretation

Dysarthric Voice Pattern

Elevated jitter and shimmer suggest motor speech involvement consistent with hypokinetic dysarthria.

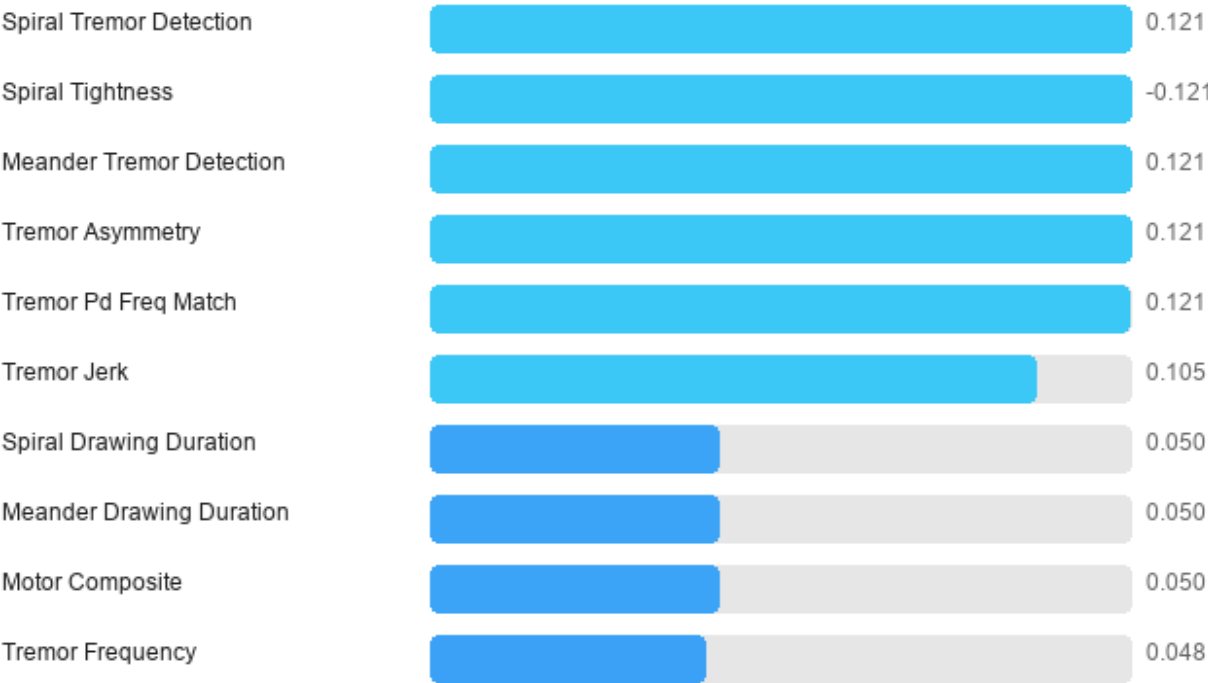
6. Motor Functions - Detailed Analysis

Health Score: 75.0% | AD Risk: 1.8% | PD Risk: 31.7% | Stage: Unilateral involvement only | Severity: low

SHAP Feature Attribution

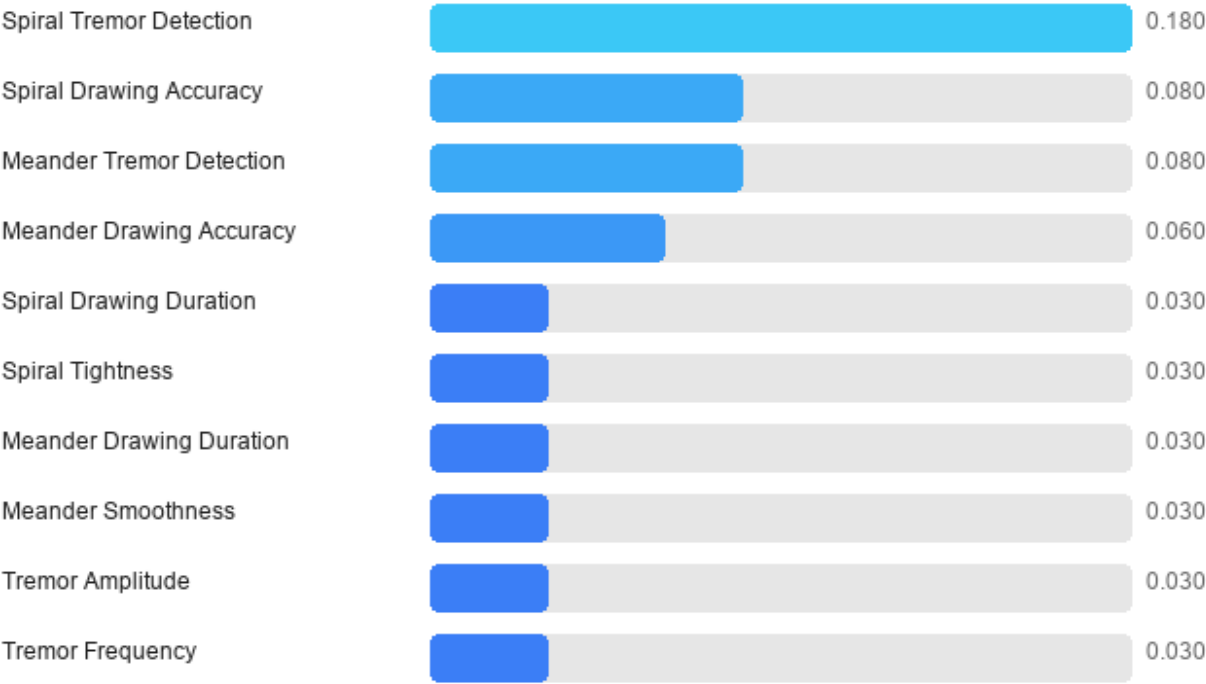
SHAP values show each feature's contribution to the risk prediction. Positive values increase risk; negative values are protective.

SHAP Feature Importance



Feature Importance Ranking

Top Features by Importance

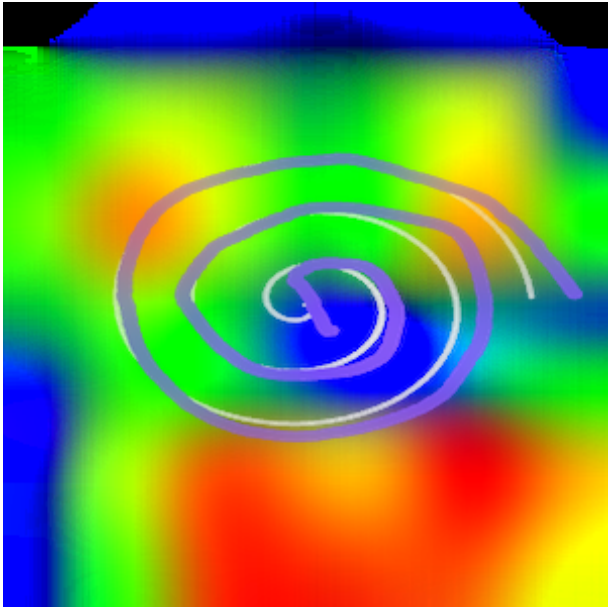


Spiral Drawing - Patient's Original

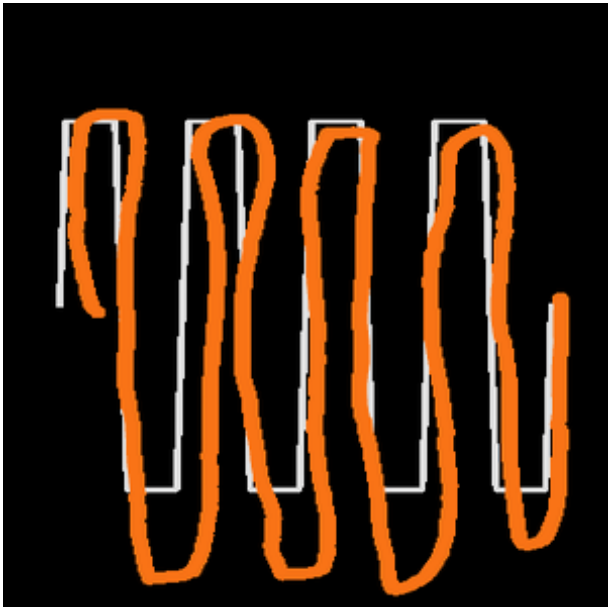


Spiral Drawing - GradCAM Heatmap Overlay

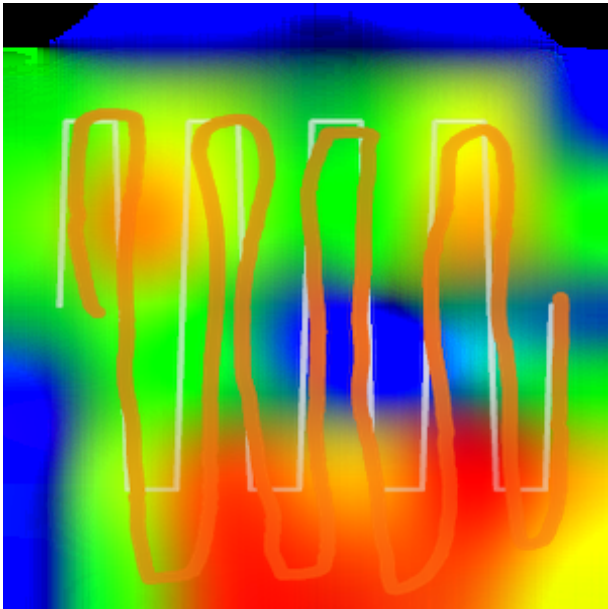
Warm regions (red/yellow) show areas the AI model focused on most when making its prediction. Cool regions (blue) had less influence.



Meander Drawing - Patient's Original

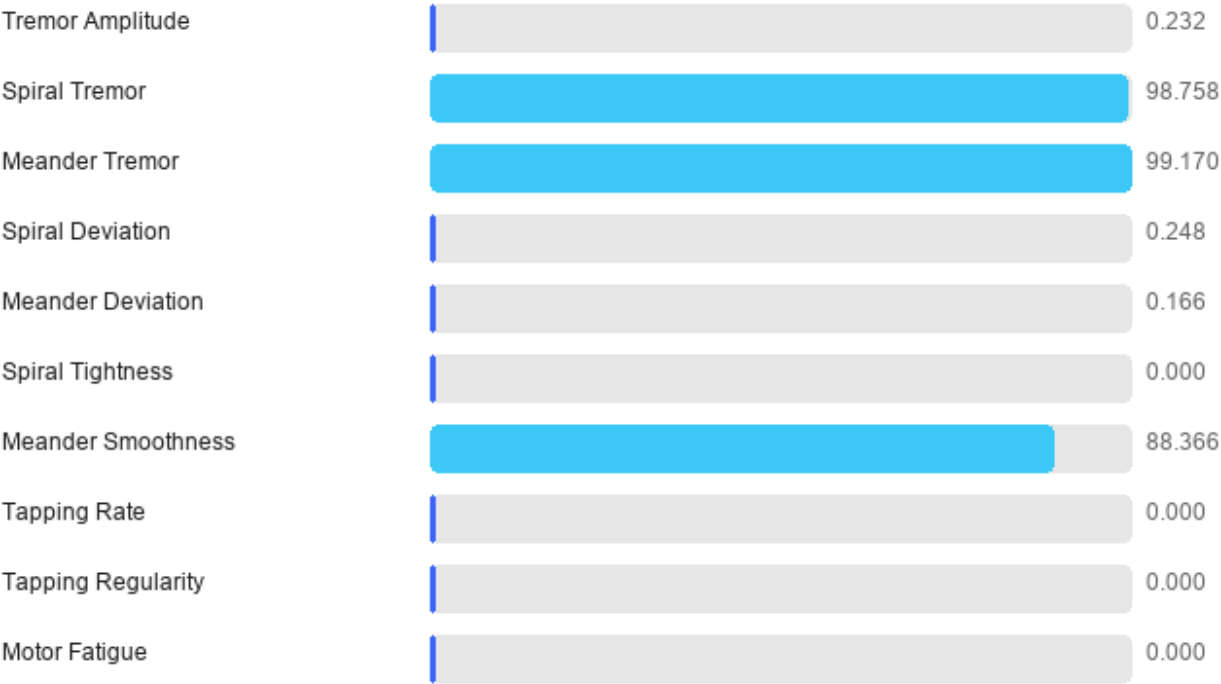


Meander Drawing - GradCAM Heatmap Overlay



Feature Analysis Bars

Motor Functions Features



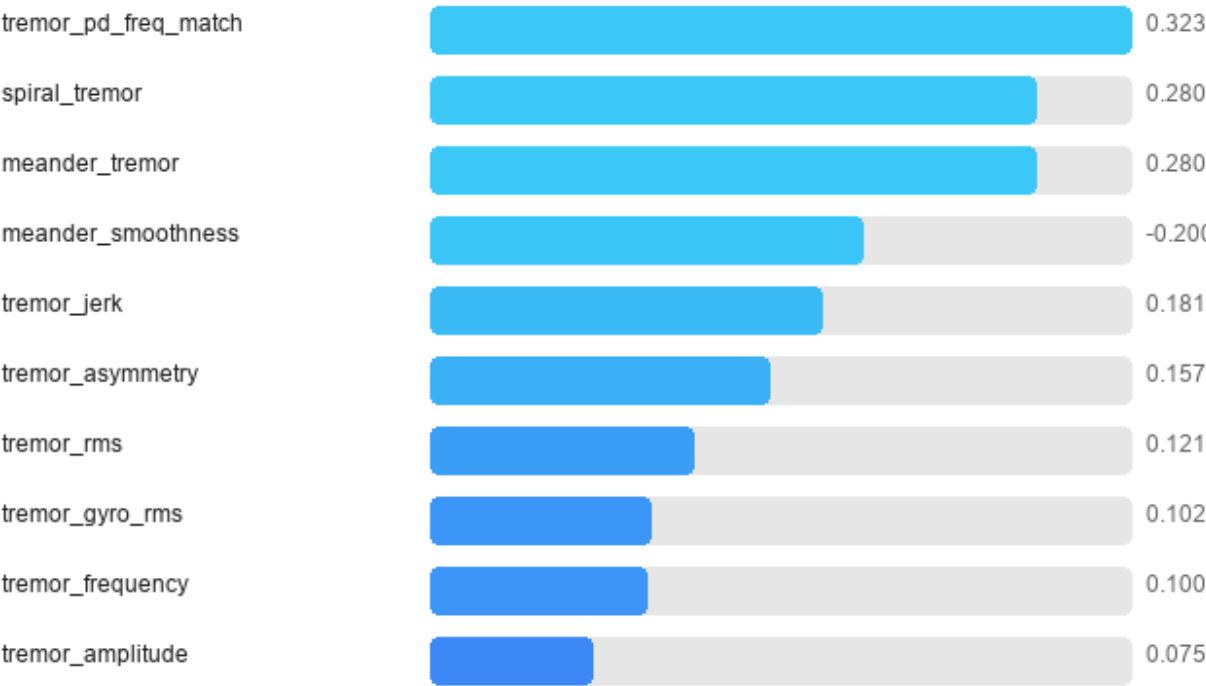
Clinical Highlights

- Tremor pattern identified in spiral drawing
- Tremor pattern identified in meander drawing
- Tremor frequency (4-6 Hz) matches Parkinsonian pattern

LIME - Local Interpretable Explanation

LIME identifies which features most influenced this specific prediction by approximating the model's decision boundary locally.

LIME Feature Weights



Integrated Gradients - Path-Integrated Attribution

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Integrated Gradient Attributions



Counterfactual Analysis - 'What-If' Scenarios

Attention Pattern Visualization

Shows where the neural network focused its attention when processing the input data.

Clinical Interpretation

Moderate PD Risk Indicators

Some motor/speech patterns (32% risk) may warrant monitoring.

Spiral Deviation - Below Expected

Spiral Deviation is 0.25, significantly below the normal range (0-0.3).

Spiral Tightness - Below Expected

Spiral Tightness is 0.00, significantly below the normal range (0.6-1.0).

Meander Deviation - Below Expected

Meander Deviation is 0.17, significantly below the normal range (0-0.3).

Tremor Amplitude - Below Expected

Tremor Amplitude is 0.23, significantly below the normal range (0-0.3).

Consistent Tremor Across Drawings

Tremor detected in both spiral and meander drawings, increasing confidence in motor dysfunction.

7. Facial Analysis - Detailed Analysis

Health Score: 93.3% | AD Risk: 1.9% | PD Risk: 22.4% | Stage: Normal | Severity: low

SHAP Feature Attribution

SHAP values show each feature's contribution to the risk prediction. Positive values increase risk; negative values are protective.

SHAP Feature Importance



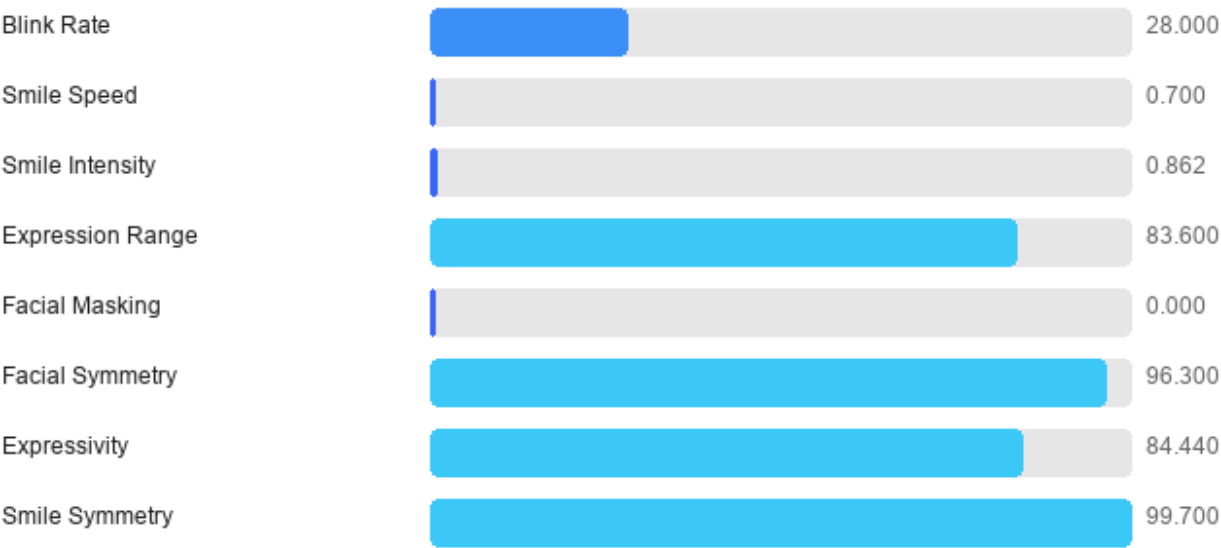
Feature Importance Ranking

Top Features by Importance



Feature Analysis Bars

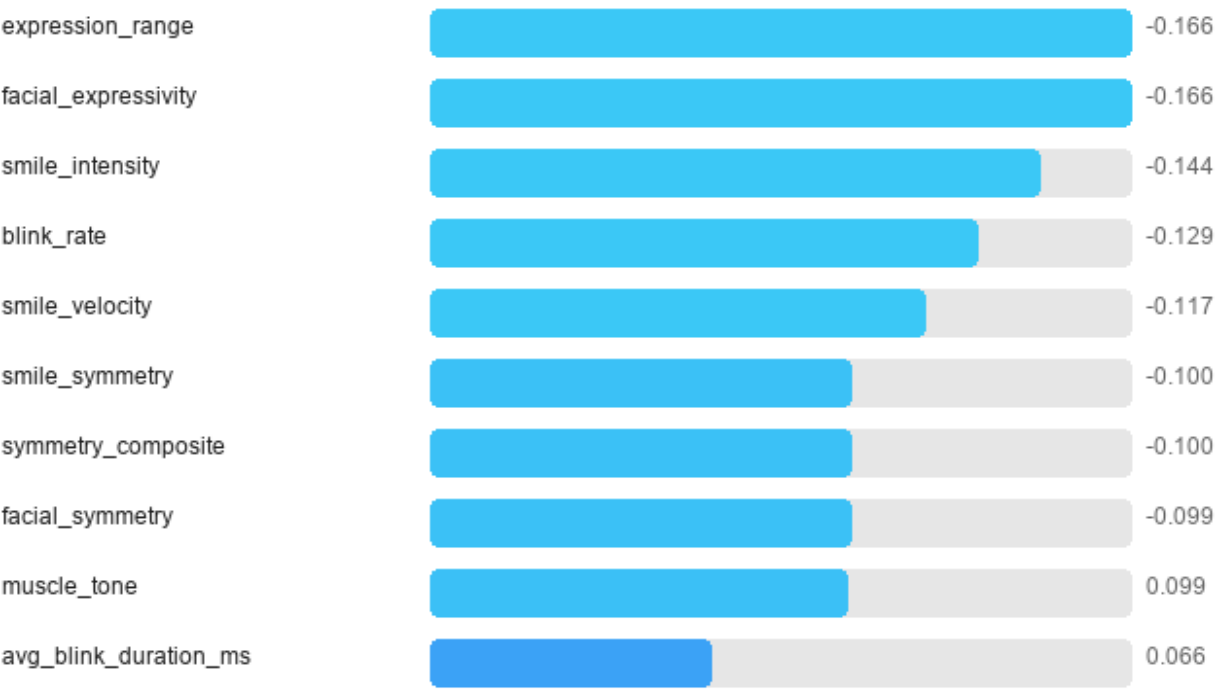
Facial Analysis Features



LIME - Local Interpretable Explanation

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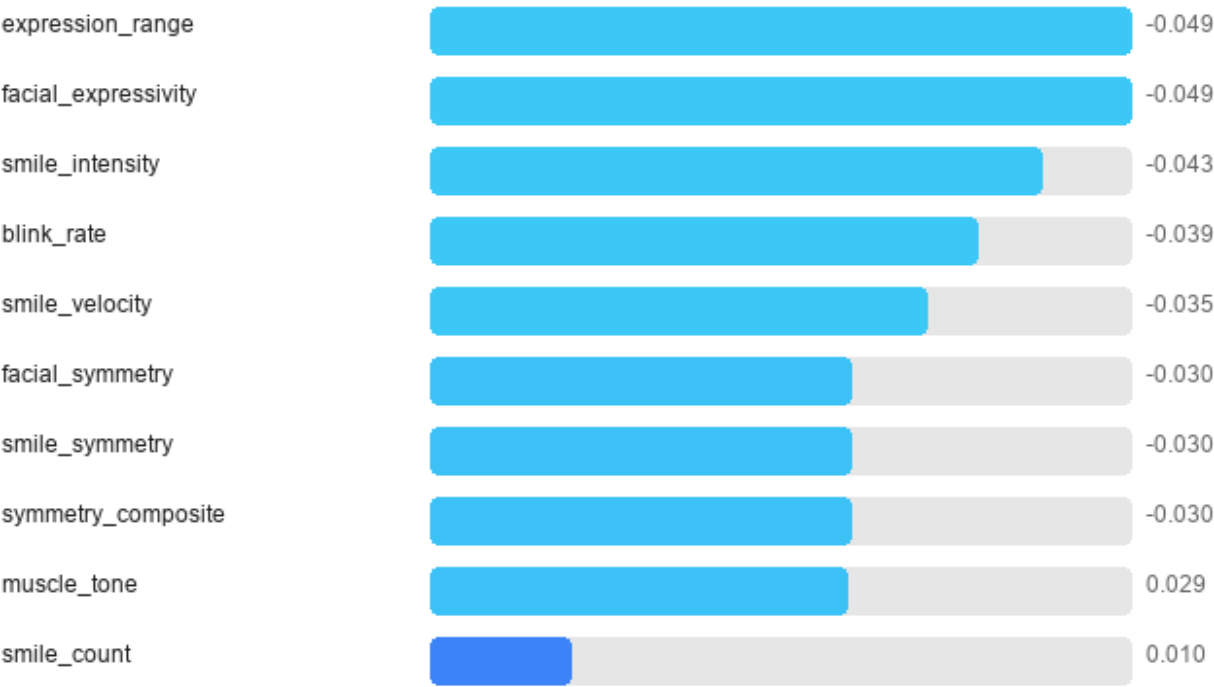
LIME Feature Weights



Integrated Gradients - Path-Integrated Attribution

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Integrated Gradient Attributions



Counterfactual Analysis - 'What-If' Scenarios

Clinical Interpretation

Facial Masking Score - Below Expected

Facial Masking Score is 0.00, significantly below the normal range (0-30).

Healthy Performance

Your facial assessment shows patterns within the normal range (score: 93/100).

8. Test Sessions Summary

#	Category	Status	Date	Score
1	motor	completed	2026-04-05	75.0%
2	motor	completed	2026-04-05	75.0%
3	motor	completed	2026-04-05	75.0%
4	facial	completed	2026-04-05	93.3%
5	facial	completed	2026-04-05	93.3%
6	facial	completed	2026-04-05	93.3%
7	facial	completed	2026-04-05	93.3%
8	facial	completed	2026-04-05	93.3%
9	facial	completed	2026-04-05	93.3%
10	facial	completed	2026-04-05	93.3%
11	motor	completed	2026-03-24	75.0%
12	facial	completed	2026-03-23	93.3%
13	facial	completed	2026-03-22	93.3%
14	facial	completed	2026-03-21	93.3%
15	facial	completed	2026-03-21	93.3%
16	cognitive	completed	2026-03-20	47.4%
17	motor	completed	2026-03-19	75.0%
18	cognitive	completed	2026-03-19	47.4%
19	motor	completed	2026-03-19	75.0%
20	cognitive	completed	2026-03-19	47.4%

9. Doctor's Notes & Recommendations

Clinical Recommendations

- 1. Current assessment scores are within normal range.
- 2. Continue routine monitoring and healthy lifestyle practices.
- 3. Regular physical exercise, cognitive engagement, and social interaction are advised.

Disclaimer: This report is generated by AI-assisted analysis and is intended as a screening tool only. It is not a definitive diagnosis. All findings should be reviewed and interpreted by a qualified healthcare professional. Clinical decisions should not be based solely on this report. NeuroVerse is a research tool and does not replace standard clinical assessment protocols.